

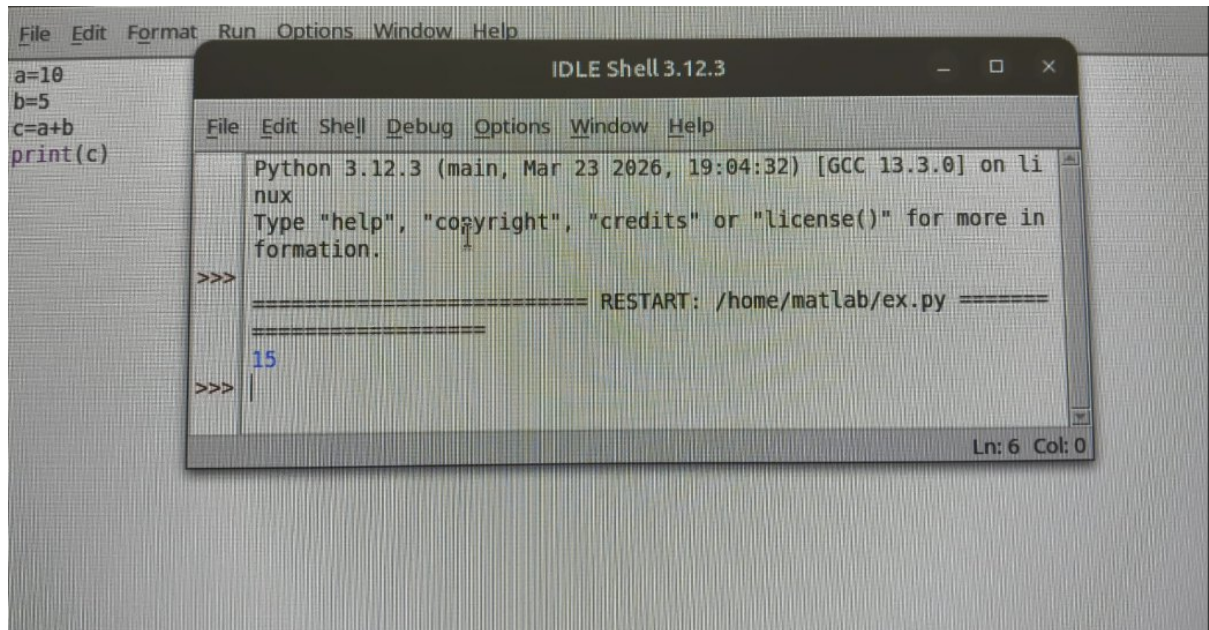
EXPLORTORY DATA ANALYSIS

PYTHON BASIC PROBLEMS

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1)



The screenshot shows the Python IDLE Shell 3.12.3 interface. The main window contains the following code:

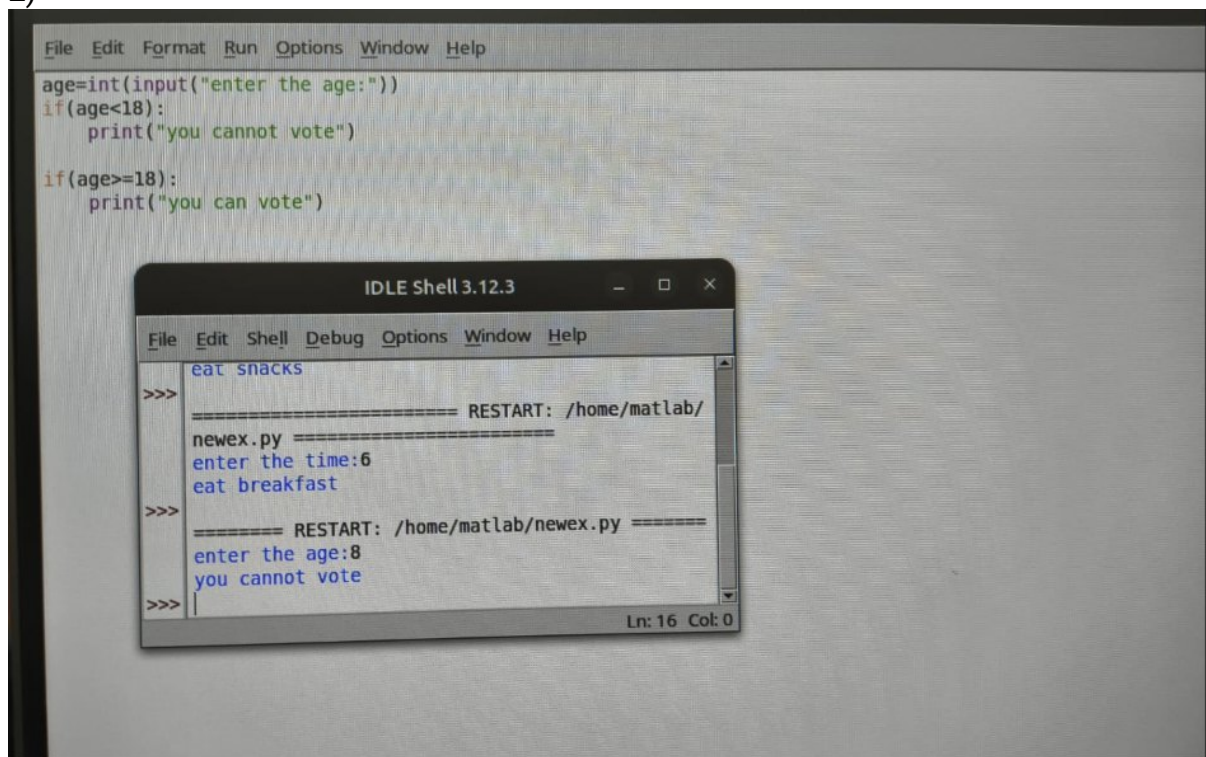
```
a=10
b=5
c=a+b
print(c)
```

The shell window shows the output of the program:

```
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on li
nux
Type "help", "copyright", "credits" or "license()" for more in
formation.
>>>
===== RESTART: /home/matlab/ex.py =====
15
>>>
```

The status bar at the bottom right of the shell window indicates "Ln: 6 Col: 0".

2)



The screenshot shows the Python IDLE Shell 3.12.3 interface. The main window contains the following code:

```
age=int(input("enter the age:"))
if(age<18):
    print("you cannot vote")
if(age>=18):
    print("you can vote")
```

The shell window shows the output of the program:

```
eat snacks
>>>
===== RESTART: /home/matlab/
newex.py =====
enter the time:6
eat breakfast
>>>
===== RESTART: /home/matlab/newex.py =====
enter the age:8
you cannot vote
>>>
```

The status bar at the bottom right of the shell window indicates "Ln: 16 Col: 0".

3)

The screenshot shows a Python IDE with a script and its execution output. The script is as follows:

```
time=int(input("enter the time:"))
if(time<9):
    print("eat breakfast")

if(time>12 and time<15):
    print("eat lunch")

if(time>18 and time<21):
    print("eat dinner")
```

The IDE Shell window shows the execution of the script:

```
newex.py ===== RESTART: /home/matlab/
enter the time:6
eat breakfast
>>>
```

The shell window also displays the file path: `Ln: 12 Col: 0`.

4)

The screenshot shows a Python IDE with a script and its execution output. The script is as follows:

```
age=int(input("enter the age:"))
if(age<18):
    print("you cannot vote")

if(age>=18):
    print("you can vote")
```

The IDE Shell window shows the execution of the script:

```
eat snacks
>>>
===== RESTART: /home/matlab/
newex.py =====
enter the time:6
eat breakfast
>>>
===== RESTART: /home/matlab/newex.py =====
enter the age:8
you cannot vote
>>>
```

The shell window also displays the file path: `Ln: 16 Col: 0`.

5)

The screenshot shows the Python IDLE 3.12.3 environment. The main editor window contains a Python script for calculating the factorial of a number. The code is as follows:

```
File Edit Format Run Options Window Help
n=int(input("enter the number"))
b=1;
for i in range(1,n+1):
    b=b*i
print("the factorial is",b)
```

An "IDLE Shell 3.12.3" window is open in the foreground, showing the execution of the script. It displays a restart message, the input "5", and the output "the factorial is 120".

```
===== RESTART: /home/matlab/newex.py
=====
enter the number5
the factorial is 120
>>>
```

6)

The screenshot shows the Python IDLE 3.12.3 environment. The main editor window contains a Python script for creating and displaying an array. The code is as follows:

```
File Edit Format Run Options Window Help
ls=[]
n=int(input("no of elements"))
for i in range(0,n):
    element=input("the element")
    ls.append(element)

print("the array is")
for i in range(0,n):
    print(ls[i])
```

An "IDLE Shell 3.12.3" window is open in the foreground, showing the execution of the script. It displays the input "5" for the number of elements, followed by five prompts for elements, and then the output of the array.

```
no of elements5
the element1
the element2
the element3
the element4
the element5
the array is
1
2
3
4
5
```

7)

```
newex.py - /home/matlab/newex.py (3.12.3)
File Edit Format Run Options Window Help

ls=[]
x=0;
n=int(input("no of elements"))
for i in range(0,n):
    element=input("the element")
    ls.append(int(element))

print("the array sum is")
for i in range(0,n):
    n=ls[i]+n

print(n)
```

IDLE Shell 3.12.3

```
File Edit Shell Debug Options Window Help

no of elements7
the element6
the element4
the element4
the element9
the element8
the element10
the element13
the array sum is
61

>>> Ln: 138 Col: 0
```

8)

```
newex.py - /home/matlab/newex.py (3.12.3)
File Edit Format Run Options Window Help

n=int(input("enter no of hours of sleep"))
if(n<2):
    print("you are sleep deprived")
if(n>2 and n<6):
    print("insufficient sleep")
if(n>6):
    print("sufficient sleep")
```

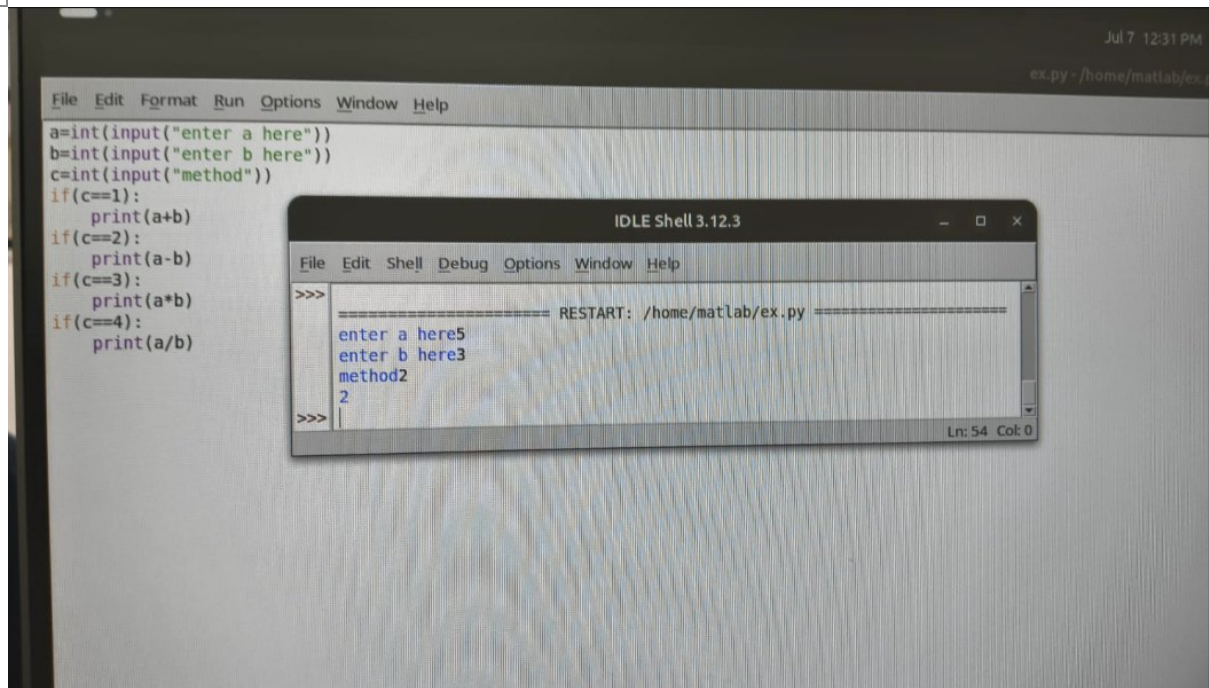
IDLE Shell 3.12.3

```
File Edit Shell Debug Options Window Help

>>>
==== RESTART: /home/matlab/newex.py ====
enter no of hours of sleep5
insufficient sleep

>>> Ln: 47 Col: 0
```

9)



10)

